On Abstract Harmonic Analysis

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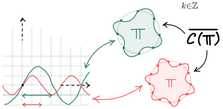
FOURIER ANALYSIS

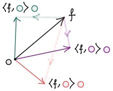
Periodic functions on $\mathbb{R}$ can be regarded as continous functions from the unit torus $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ to $\mathbb{C}$. These periodic functions form a $\mathbb{C}$-inner product space under the inner product,

$$\langle f, g \rangle := \int_0^1 f(e^{2\pi i t})\overline{g(e^{2\pi i t})}dt$$

Regarding the closure of this space, namely $\overline{C(\mathbb{T})}$ as a Hilbert space, the maps $x \mapsto e^{2\pi i k x}$ for $k \in \mathbb{Z}$ form an orthonormal basis for $\overline{C(\mathbb{T})}$. The Fourier coefficients of $f : \mathbb{T} \to \mathbb{C}$, is given by the projections of f along $x \mapsto e^{2\pi i k x}$, that is $c_k = \langle f, e_k \rangle$, where $e_k := e^{2\pi i k x}$. Thus, the Fourier series is given by

$$\mathcal{F}(f) := \sum_{k\in\mathbb{Z}} \langle f, e_k \rangle e_k$$





Surprisingly, convergence in $\overline{C(\mathbb{T})}$ does not guarantee pointwise convergence. In order to ensure this, we impose some additional conditions and arrive at the following known result.

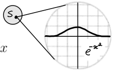
Theorem. $f \in \overline{C(\mathbb{T})}$ be continous, piecewise C^1. Then $F_n(f) \rightrightarrows f$.

The Schwartz Space is a particularly nice class of functions

$$\mathcal{S} = \{f : \mathbb{R} \to \mathbb{C} \mid f \text{ is } C^\infty, \sigma_{m,n}(f) < \infty \text{ for all integers } n, m \geq 0\}$$

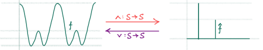
with the inner product,

$$\langle f, g \rangle := \int_{-\infty}^{\infty} f(x)\overline{g(x)}dx$$



This space consists of functions whose all derivatives are rapidly decreasing, that is, faster than any polynomial. We define the Fourier Transform on this domain. The choice of this domain is made, so that much of the proofs and results later are elegant and intuitive. **Fourier Transform.** For $f \in \mathcal{S}$, we define its Fourier transform as

$$\hat{f}(y) := \int_{-\infty}^{\infty} f(x)e^{-2\pi i x y}dx$$



The Fourier Transform maps the Schwartz Space to itself, that is, $\hat{\mathcal{S}} \subset \mathcal{S}$ and we have the Inverse Transform on the Schwartz Space as well. However without injectiveness, one cannot establish unitarity on $\overline{\mathcal{S}}$ without an isometry.

PLANCHEREL THEOREM

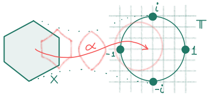
An isometry is a bijection of metric spaces that preserves distances, that is, $\|f\|_X = \|Tf\|_Y$ where $T : X \to Y$ is an isometry. The Plancherel Theorem shows $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ is indeed an isometry.

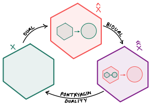
Plancherel Theorem. For $f \in \mathcal{S}$, we have $\|f\|_2 = \|\hat{f}\|_2$.

Isometries are automatically injective. Thus, the Fourier Transform is established to be a unitary map on $\overline{\mathcal{S}}$. We shall try to arrive at a more general version of this theorem for LCA Groups.

POTRYAGIN DUALITY

Topological groups are groups whose operations are continuous under some topology. We consider groups with the σ-locally compact topology, namely LCA groups. These groups are always seperable and posses an absorbing compact exhaustion, namely $X = \bigcup_n K_n$. Characters are group homomorphisms $\alpha : X \to \mathbb{T}$ that are continuous under the σ-LC topology.





Let the dual group be $\hat{X} = \{\alpha : \alpha \text{ is a character of } X\}$. Then $\hat{X}$ is a group under pointwise product. Next, we define a metric on $\hat{X}$,

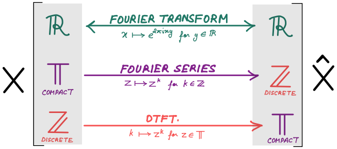
$$\hat{d}_n(\alpha, \beta) = \sup_{x\in K_n} |\alpha(x) - \beta(x)| \quad \text{and} \quad \hat{d}(\alpha, \beta) = \sum_{n=1}^{\infty} \frac{1}{2^n}\hat{d}_n(\alpha, \beta)$$

This metric, induces a σ-locally compact topology on $\hat{X}$, irrespective of the choice of K_n. Lev Pontryagin showed that

Pontryagin Duality. Between any LCA group X and its bidual, there is a canonical isomorphism $X \cong \hat{\hat{X}}$, given by the map

$$X \to \hat{\hat{X}} \quad a \mapsto \delta_a, \quad \delta_a(\chi) = \chi(a)$$

where δ_a is evaluation of the character χ at the point a. This establishes the following duality.


X
ℝ
FOURIER TRANSFORM
$x \mapsto e^{2\pi i x y}$ for $y \in \mathbb{R}$
ℝ
𝕋
COMPACT
FOURIER SERIES
$z \mapsto z^k$ for $k \in \mathbb{Z}$
ℤ
DISCRETE
$\hat{X}$
ℤ
DISCRETE
DTFT.
$k \mapsto z^k$ for $z \in \mathbb{T}$
𝕋
COMPACT

GENERAL PLANCHEREL THEOREM

For an LCA group G, the Fourier Transform is defined on $L^1(G)$, which is space of all absolutely Haar Integrable functions G to $\mathbb{C}$. The Fourier Transform is a function on the dual group, given by

$$\hat{f} \in \hat{G} \quad \text{defined as} \quad \hat{f}(\chi) = \int_G f(x)\overline{\chi(x)}dx \quad \text{for } \chi \in \hat{G}$$

We have $L^1(G) \subset l^2(G)$, which is a Hilbert space. Thus, there is already an inner product on $L^1(G)$, which induces a metric. The closure of $L^1(G)$ under this metric is denoted as $L^2(G)$. The Plancherel Theorem asserts the preservation of this norm under Fourier Transformation.

General Plancherel Theorem. There is unique unitary isomorphism

$$\mathcal{F} : L^2(G) \to L^2(\hat{G})$$

extending the Fourier Transform from the dense subspace $L^1(G) \cap L^2(G)$ to $L^2(G)$. Thus, for $f \in L^2(G)$,

$$\|f\|_{L^2} = \|\hat{f}\|_{L^2}$$

This extension is sometimes called the **Plancherel Transform** and was one of the major results proven by Michel Plancherel.

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